

Q: 3.4

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Rationalize

if $n = 8 - 3\sqrt{7}$, find $\left(n + \frac{1}{n}\right)^2$

so

$$\left(\frac{8 - 3\sqrt{7} + 1}{8 - 3\sqrt{7}} \right)^2$$

Note: we have given $n = 8 - 3\sqrt{7}$

but we don't have given $\frac{1}{n}$ so for

$$\frac{1}{n} = \frac{1}{8 - 3\sqrt{7}} \quad \therefore \text{so now we have } \frac{1}{n} \text{ also value of } \frac{1}{n}$$

so

$$\frac{1}{8 - 3\sqrt{7}} \times \frac{8 + 3\sqrt{7}}{8 + 3\sqrt{7}}$$

$$= \frac{8 + 3\sqrt{7}}{(8)^2 - (3\sqrt{7})^2} = \frac{8 + 3\sqrt{7}}{64 - 9(7)} = \frac{8 + 3\sqrt{7}}{64 - 63}$$

$$\frac{8 + 3\sqrt{7}}{1} = 8 + 3\sqrt{7} \quad \text{is the value of } \frac{1}{n}$$

Now we have already given the value of n and then we have already find value of $\frac{1}{n}$ so

Now we can easily find $(n + \frac{1}{n})^2$ value
so

$$n + \frac{1}{n} = \frac{\cancel{8-3\sqrt{7}} + \cancel{1}}{(8-3\sqrt{7}) + (8+3\sqrt{7})}$$

$$8-3\sqrt{7} + 8+3\sqrt{7}$$

$$(n + \frac{1}{n}) = \boxed{16}$$

Now taking S.R.B.S

$$(n + \frac{1}{n})^2 = (16)^2$$

$$(n + \frac{1}{n})^2 = 256$$

Ans.